# MLPR Lab 11

## INSTRUCTIONS

This assignment builds upon the last week's lab. The neural network that you will build this week will be able to solve XOR problem which a single perceptron was not be able to do since it could only classify linearly separable data. Additionally, you will also learn to use a few hyperparameters in this assignment that will help you train a neural network faster and more efficiently.

- Using Google Colab is recommended for this lab as there might some issues due to tensorflow installation and their dependencies.
- There can be a difference in values in the plots as compared to the given output to what you produce.

Step 1: Import libraries

- Numpy
- Keras from tensorflow
- Dense layers from keras
- Matplotlib

Step 2: Take XOR input data and store in one variable (Input data), Store output data of XOR in another variable (Target data)

Step 3: Create the model.

- Define sequential model.
- Add first layer in the model as given parameters
    - Input data=2
    - No of node =8
    - Activation = relu
- Add the second layer
    - No of node=1
    - Activation=sigmoid
- Keep learning rate = 0.1
- Use SGD as an optimizer with given learning rate.

Step 4: Compile the model with defined optimizer in previous step with MSE as loss calculator.

Step 5: Now, we need to record the learning rates so that we can capture the no of epochs at model converging.

- Create a callback to record learning rates using ***keras.callbacks.Callback***.

Step 5: Train the model and monitor the convergence and learning rates.

- Converged=False
- Target_loss=0.001  (Convergence criteria)
- Losses =[]
- Learning_rates=[]

Step6 : Initiate the while loop to check if the network is not converged. Check this after every 5 or 10 epochs. If network is not converging at this particular epoch, use this-

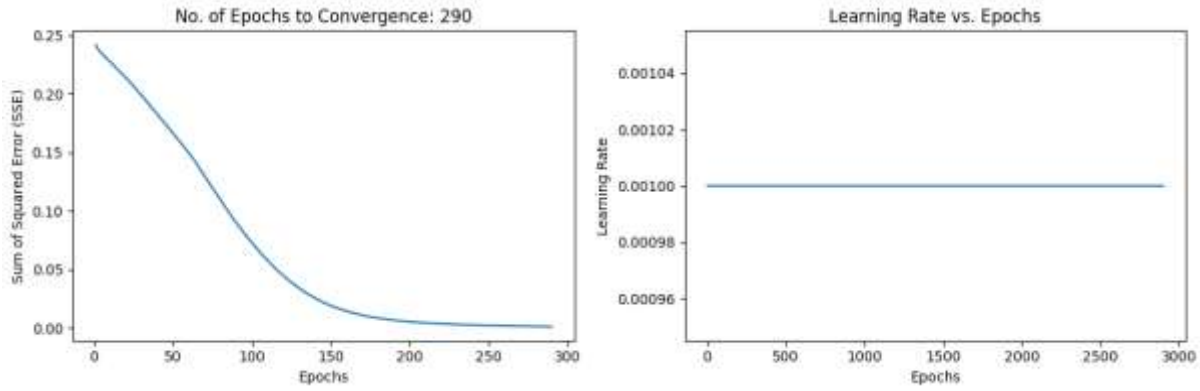
History =model.fit(X,y, epoch=10, verbose=1, callbacks =[LearningrateCallback()])

Loss=history.history['loss'][0]

Losses.append(loss)

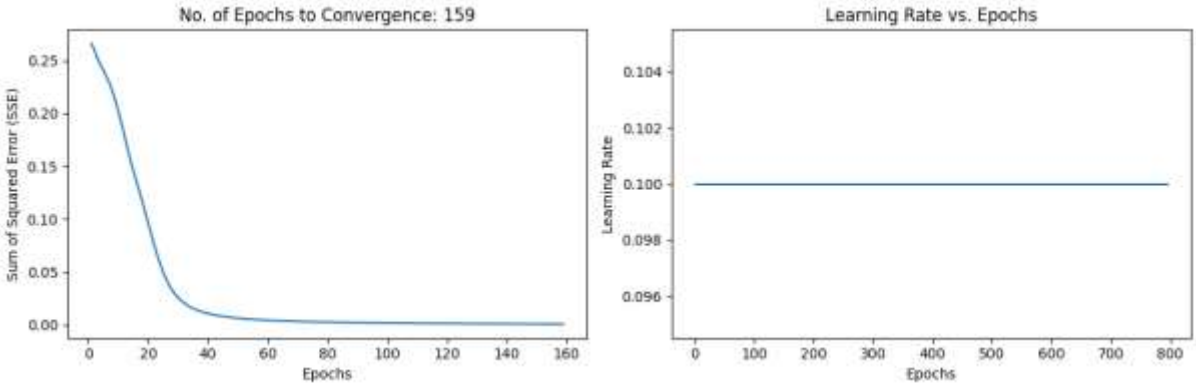
Step6: Plot the SSE (Sum of squared error) vs. Number of epochs. In title, it should show the no of epoch at which the network converges. See the reference output.

Step7: Plot the graph of learning rate vs Number of epochs. See the reference output.



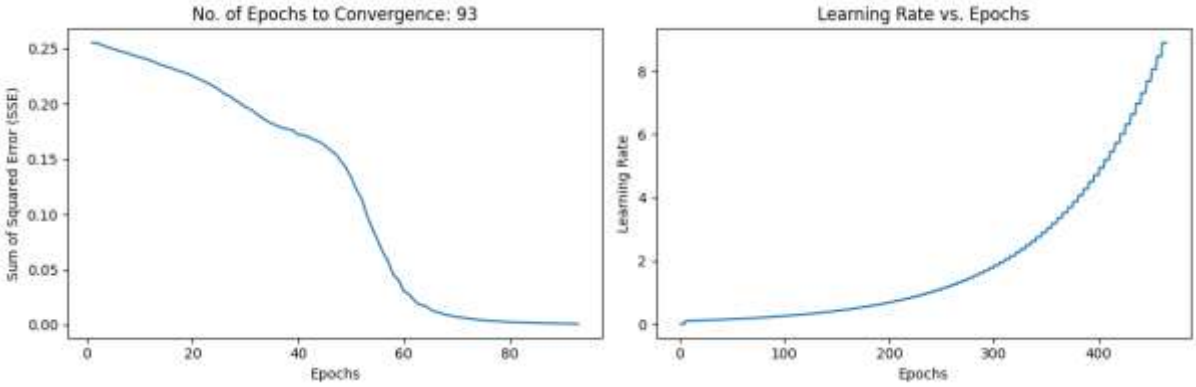

Step8: Now you need to do the same task by incorporating momentum-based learning rate to accelerate the learning.

- Keep all the parameters same as above for all the steps.
- In step3, Add the momentum = 0.9 and use this momentum in optimizer along with the given learning rate.
- Keep the rest as same and generate both the graphs shown below.



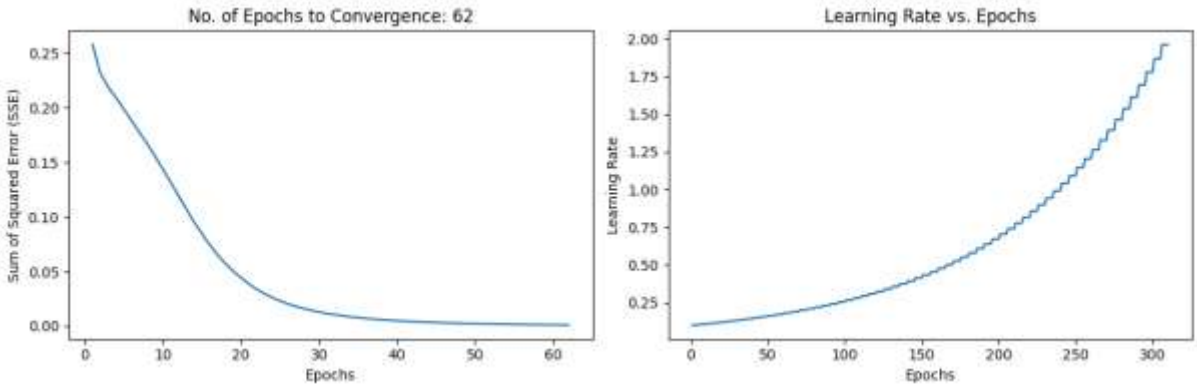

Step9: Here, we use adaptive based learning rate. Adaption criteria of learning rate is given below (Taken from LEc-17, slide 14).

- If SSE at the current epoch exceeds the previous by more than 1.04, then decreased the learning rate by 0.7. If error is less than the previous one, increase the learning rate by 1.05.
- Train the model again with new learning rate . Do this after every 5 epochs.
- Do not use momentum here. Just replicate step1 to step 7 and produce the graphs.




Step 10: Now use both Momentum and adaptive based learning rate.

- For this, keep step 9 as it is and just add momentum=0.9 in optimizer along with the learning rate



# REPORT QUESTIONS

1. How does a multilayer neural network solve the XOR problem?
2. In a backpropagation neural network, why are the middle layers called “hidden”?
3. How are hyperparameters different from model parameters?
4. Explain the concept of accelerated learning in backpropagation.
5. What are two heuristics applied in adaptive learning rate?

## **DELIVERABLES: Items to be uploaded on LMS-**

Main code file along with the complete code, all the four graphs and the report answers in one file in PDF format.